Maithili Vyas

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Professional Summary

Aspiring Game Developer with a strong foundation in Unity, C#, and game design. Passionate about creating interactive gaming experiences and exploring AI integration in games. Experienced in Virtual Production and Interactive 3D Environments through practical exposure to Unreal Engine and Unity.

Education

JIET University — **Jodhpur**
B.Tech in Computer Science (AI & ML), CGPA: 9.34 — *2023 – Present*
Currently in 6th Semester.

Experience

InKrid Studios
Game Development Intern — *Summer 2024*

- Gained hands-on experience in real-time technologies including Unreal Engine 5 and Unity.
- Focused on environment creation, 3D level design, and game prototyping.
- Developed understanding of the complete game development lifecycle from ideation to deployment.
- Collaborated on cinematics and interactive virtual reality experiences.

Technical Skills

- **Game Engines:** Unreal Engine 5 (Blueprints, Level Design), Unity (C# scripting).
- **Development:** VR/AR Development, 3D Environment Design, Lighting & Rendering.
- **Tools:** Visual Studio, Version Control.

Projects

Unity & Mobile Development

Augmented Reality (AR) Application — *Unity*

- Developed an interactive AR application using Unity and AR Foundation, implementing robust plane detection and raycasting for precise digital asset placement in the physical world.
- Integrated interactive UI elements that allow users to manipulate 3D objects (scale, rotate, translate) in real-time using touch gestures on mobile devices.
- Optimized high-fidelity 3D models and textures for mobile performance, ensuring a stable frame rate on Android devices while maintaining visual quality.

2D Dodge Game *Unity*

- Engineered a fast-paced survival game utilizing Unity's Rigidbody2D physics engine for fluid player movement and responsive collision detection.
- Implemented an object pooling system for obstacle generation to manage memory efficiency and performance during high-intensity gameplay sequences.
- Designed a progressive difficulty algorithm that dynamically increases the speed and frequency of obstacles based on the player's survival time, keeping the gameplay engaging.

Unreal Engine & Virtual Reality

Virtual Reality Museum Experience *Unreal Engine 5*

- Architected a fully immersive VR museum environment for the Meta Quest, utilizing modular assets to construct a cohesive and spatially intuitive layout.
- Programmed complex interaction mechanics using Blueprints, enabling users to physically grab, inspect, and manipulate museum artifacts with realistic hand-tracking physics.
- Implemented proximity-based logic systems, such as automatic sliding doors and dynamic audio zones that trigger ambient music and narration as the user navigates specific exhibits.

Ancient Temple Cinematic *Unreal Engine 5*

- Directed and produced a cinematic sequence featuring a Nataraja statue, utilizing Unreal Engine's Sequencer for precise camera cuts, focal length adjustments, and depth of field control.
- Mastered environmental storytelling by designing a detailed cave ecosystem, using volumetric fog and global illumination (Lumen) to create dramatic light shafts and atmosphere.
- Integrated high-quality audio layers, syncing sound effects and background scores to visual cues to enhance the mystical ambiance of the scene.

Japan Station Cinematic *Unreal Engine 5*

- Constructed a photorealistic night scene inspired by Japanese railway stations, using advanced material instances to simulate wet asphalt, reflective puddles, and neon signage.
- Utilized a combination of stationary and dynamic lighting techniques, including reflection captures and rect lights, to achieve a moody, cyberpunk-inspired aesthetic.
- Enhanced the scene with dynamic particle systems for rain and thunder, utilizing Niagara visual effects to create a living, breathing storm environment.

FPS Shooter Game *Unreal Engine Blueprints*

- Developed a robust First-Person Shooter framework featuring a custom character controller, weapon sway, and recoil mechanics for satisfying gunplay.
- Programmed AI agents using Behavior Trees and Blackboard systems, enabling enemies to patrol, detect the player via sight/sound perception, and engage in tactical combat.
- Designed a comprehensive UI/HUD system using UMG widgets to display real-time health, ammo count, and a dynamic "Game Over" screen upon player death.

Lock-and-Key Mini Game *Unreal Engine Blueprints*

- Created a logic-driven puzzle game requiring players to explore a level, locate scattered keys, and unlock a final gateway, emphasizing exploration and spatial awareness.
- Implemented a NavMesh-based AI pursuer that dynamically tracks the player's location, adding an element of tension and urgency to the gameplay loop.
- Scripted visual feedback systems, such as glowing emissive materials on keys and door animations, to clearly communicate game state changes to the player.

Extracurricular Activities

- **College Newsletter Team:** Active member of the editorial team, contributing to content creation and publication.